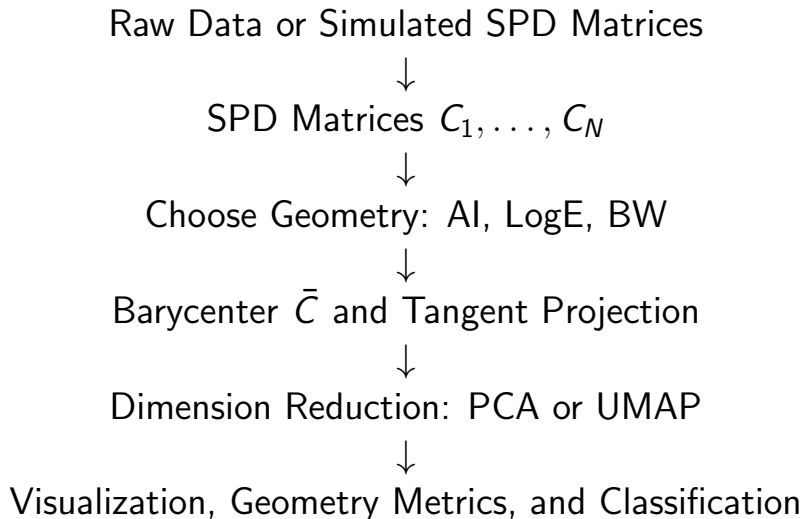


Overall Project Pipeline



Dimension Reduction and Manifold Learning

- ▶ Many datasets exist in very high-dimensional spaces, making visualization and learning difficult.
- ▶ **Dimension reduction** attempts to represent the data in a lower-dimensional space while preserving important structure.
- ▶ **Linear methods** such as PCA assume the data lie near a linear subspace.
- ▶ **Manifold learning methods** such as UMAP assume the data may lie on a curved nonlinear manifold and attempt to preserve local geometric structure and neighborhoods.
- ▶ In this project, there are two different manifold ideas:
 1. The original SPD matrices already lie on a Riemannian manifold defined by AI, LogE, or BW geometry.
 2. After tangent projection, we apply manifold learning methods such as UMAP to study whether the tangent features themselves have lower-dimensional nonlinear structure.

Dimension Reduction Methods

▶ PCA

- ▶ linear dimension reduction
- ▶ useful for studying global variance structure
- ▶ gives interpretable variance explained and loadings

▶ UMAP

- ▶ nonlinear dimension reduction
- ▶ useful for studying local neighborhood structure
- ▶ can be run on tangent features or on precomputed manifold distances

▶ More options

- ▶ Isomap
- ▶ Spectral embedding
- ▶ Autoencoder

Task 1: Compare Manifold UMAP vs Tangent-Space UMAP

A. Manifold UMAP

$$D_{ij}^{(g)} = d_g(C_i, C_j), \quad g \in \{AI, LogE, BW\}$$

- ▶ Compute pairwise SPD distances.
- ▶ Run `UMAP(metric="precomputed")`.
- ▶ This asks what structure is present directly in the manifold distances.

B. Tangent-Space UMAP

$$X_i^{(g)} = \text{vec} \left(\log_{\bar{C}_g}^{(g)}(C_i) \right)$$

- ▶ Compute the metric-specific barycenter.
- ▶ Project matrices to tangent space.
- ▶ Run `UMAP(metric="euclidean")` on tangent vectors.

Question: Does tangent projection preserve the same structure as the original manifold distances?

Task 1: What to Plot and Report

1. For each geometry $g \in \{AI, LogE, BW\}$, make two UMAP plots:
 - ▶ manifold-distance UMAP
 - ▶ tangent-space UMAP
2. Color each point by class label or group.

Geometry	Manifold UMAP	Tangent UMAP
AI	plot	plot
LogE	plot	plot
BW	plot	plot

Metrics for Task 1

► **Neighborhood Overlap**

Measures how many nearest neighbors from the original manifold distances are preserved after tangent projection.

Python: `sklearn.neighbors.NearestNeighbors`

► **Trustworthiness**

Measures whether points that appear close in the embedding were also close in the original space.

Python: `sklearn.manifold.trustworthiness`

► **Spearman Distance Correlation**

Measures whether pairwise distance rankings are preserved between manifold distances and tangent-space distances.

Python: `scipy.stats.spearmanr`

Task 2: Dimension Reduction for Classification

- ▶ Build three classification pipelines for each geometry: AI, LogE, and BW.

- ▶ **Pipeline A: Full tangent features**

$$C_i \rightarrow X_i^{(g)} \rightarrow \text{Classifier}$$

- ▶ **Pipeline B: PCA-reduced tangent features**

$$C_i \rightarrow X_i^{(g)} \rightarrow \text{PCA}_q(X_i^{(g)}) \rightarrow \text{Classifier}$$

- ▶ **Pipeline C: UMAP-reduced tangent features**

$$C_i \rightarrow X_i^{(g)} \rightarrow \text{UMAP}_q(X_i^{(g)}) \rightarrow \text{Classifier}$$

- ▶ Classifiers: k -nearest neighbors, logistic regression, SVM, etc.

Task 2: Classification Comparisons

1. Use the same train/test splits for AI, LogE, and BW.
2. Compare latent dimensions:

$$q \in \{5, 10, 20, \dots\}$$

3. Fit dimension reduction only on the training set.
4. Transform validation/test observations using the fitted PCA or UMAP object.
5. Report a table of classification metrics:
 - ▶ Accuracy
 - ▶ Standard Deviation
 - ▶ Precision
 - ▶ Recall
 - ▶ AUC
 - ▶ Runtime

Conceptual Algorithm: Nested CV Classification

Outer loop: 5-fold cross-validation

1. Split the data into 5 outer folds.
2. For each outer fold:
 - 2.1 Hold out one fold as the test set.
 - 2.2 Use the remaining four folds as the training set.
 - 2.3 Compute the barycenter using only the training SPD matrices.
 - 2.4 Project training and test matrices to tangent space using the training barycenter.
 - 2.5 Half-vectorize (i.e. flatten matrix and take only upper or lower triangle)
 - ▶ *If standardizing: standardize train matrices and fit test matrices using training standarization*
 - 2.6 Tune dimension reduction and classifier parameters using inner CV on the training set only.
 - 2.7 Refit the selected pipeline on the full outer-training set.
 - 2.8 Evaluate on the held-out outer-test set.
3. Average performance across the 5 outer folds.

Task 3: Raw vs Standardized Tangent Features

Question 1: Do certain metrics naturally produce better-scaled tangent features?

- ▶ Compute the standard deviation of each tangent feature:

$$s_j = \text{sd}(x_j)$$

where x_j is the j -th tangent-space feature across all samples.

- ▶ Create boxplots or line plots of feature standard deviations for AI, LogE, and BW.

Task 3: Raw vs Standardized Tangent Features

Question 2: Does standardization change the latent structure?

- ▶ Standardize each tangent feature:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

- ▶ Run PCA on both raw and standardized tangent features.
- ▶ Plot cumulative variance explained:

$$\text{CVE}(q) = \frac{\sum_{j=1}^q \lambda_j}{\sum_{j=1}^d \lambda_j}$$

Task 3: Raw vs Standardized Tangent Features

Question 3: Does standardization affect classification performance?

- ▶ Compare accuracy using:
 - ▶ raw tangent features
 - ▶ standardized tangent features
- ▶ Create grouped barplots for AI, LogE, and BW.

Datasets for the Project

- ▶ **Synthetic SPD matrices**
 - ▶ controlled class structure
 - ▶ known source of variation
 - ▶ fast enough to rerun many times
- ▶ **ETH-80 image covariance descriptors**
 - ▶ one covariance matrix per image
 - ▶ use image features such as position, color, and gradients
- ▶ **EEG / BCI or fMRI**
 - ▶ one covariance matrix per EEG trial
 - ▶ one covariance matrix per brain scan

